



Penrith Selective
High School

2011

Higher School Certificate
Trial Examination

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided
- All necessary working should be shown in every question
- All questions are of equal value
- Staple this test to your answers
- Attempt Questions 1 – 7

Total marks – 84

Question	Mark
1	/12
2	/12
3	/12
4	/12
5	/12
6	/12
7	/12
Total	/84
Percentage	

Student's Name: _____

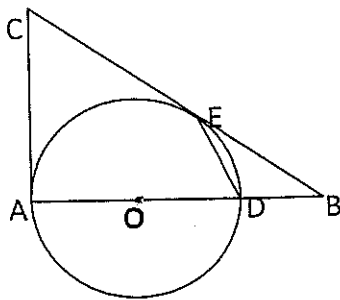
Teacher: _____

Question 1 (Start a new page).**(12 marks).**

- a) Find the coordinates of the point which divides the interval joining $(-3,5)$ to $(8,4)$ in the ratio 1:3 (2)
- b) Find the equations of the two lines inclined at 45° to the line $3x + y = 8$ and passing through the point $A(4,0)$. (3)
- c) Find $\int x\sqrt{3-x}dx$, using the substitution $u = 3 - x$ (3)
- d) Solve $\frac{2x+5}{x} \geq 1$ (2)
- e) Without using a calculator, show that $\sin \left[\cos^{-1} \left(\frac{3}{5} \right) + \tan^{-1} \left(-\frac{3}{4} \right) \right] = \frac{7}{25}$ (2)

Question 2 (Start a new page).**(12 marks).**

- a) Find the Cartesian equation of the curve and describe it geometrically if $x = \sin \theta + \cos \theta$ and $y = \sin \theta - \cos \theta$ (3)
- b) i) Derive the equation of the normal to $x^2 = 4ay$ at $(2ap, ap^2)$. (3)
- ii) The chord joining $P(2ap, ap^2)$ to $Q(2aq, aq^2)$ passes through $T(0, -2a)$ show that $pq=2$. (3)
- c) AD is a diameter of the circle centre O . E is a point on the circumference. Tangents at A and E meet at C . CE and AD meet at B . Prove that angle $ACB = 2(\text{angle } DEB)$. (3)



Question 3 (Start a new page).**(12 marks).**

The polynomial $P(x) = 0$ has a double root at $x = a$.

- a) By putting $P(x) = (x - a)^2 Q(x)$, show that $P'(a) = 0$. (2)
- b) The equation $P(x) = mx^4 + nx^3 - 6x^2 + 22x - 12 = 0$ has a double root at $x = 1$. Find the values of m and n . (2)
- c) Using the fact that $x = 3$ is also a root of $P(x)$, express $P(x)$ in factorised form. (2)
- d) Solve $P'(x) = 0$ and state if $x = 1$ represents a local minimum or local maximum. (2)
- e) Show on a sketch of $P(x)$ where it cuts the axes and the x -values of the turning points. (3)
- f) From the graph solve $P(x) > 0$ (1)

Question 4 (Start a new page).**(12 marks).**

- a) Let $S_n = 1^2 + 2^2 + \dots + n^2$, for $n = 1, 2, 3, \dots$
 - i) Use Mathematical Induction to prove that for $n = 1, 2, 3, \dots$
$$S_n = \frac{1}{6}n(n+1)(2n+1)$$
 (4)
 - ii) By using the result of part (i) estimate the least n such that
 $S_n \geq 10^{11}$ (2)
- b)
 - i) How many different arrangements are there of the letters
ARRANGEMENT (1)
 - ii) How many of these begin with the letter R? (2)
- c) Evaluate $\int_0^{\frac{\pi}{4}} \tan^3 x \sec^4 x \, dx$, using the substitution $u = \tan x$ (3)

Question 5 (Start a new page).**(12 marks).**

- a) Evaluate $\int_0^{\frac{\pi}{2}} \cos^2 3x \, dx$ (3)
- b) i) Express $-\sin x - \sqrt{3}\cos x$ in the form $R\cos(x - \alpha)$ (3)
- ii) Hence solve $-\sin x - \sqrt{3}\cos x = 2$ for $0 \leq x \leq 2\pi$. (3)
Give the solutions correct to 3 decimal places.
- c) Prove that $\frac{1 - \tan\theta \tan 2\theta}{1 + \tan\theta \tan 2\theta} = 4\cos^2\theta - 3$ (3)

Question 6 (Start a new page).**(12 marks).**

- a) $f(x) = x - \frac{1}{x}, x > 0$
- i) Show that $f(x)$ has no stationary points. (1)
- ii) Describe the behaviour of $f(x)$ as x approaches the extremities of its domain. (1)
- iii) Sketch, on one diagram, graphs of $y = x$, $y = f(x)$ and $y = f^{-1}(x)$ (3)
- iv) If $x = y - \frac{1}{y}$ and $y > 0$, simplify, in terms of y , $x + \sqrt{x^2 + 4}$ (2)
- v) Write the expression for $f^{-1}(x)$ in terms of x (1)
- b) i) Find $\frac{d}{dx}[\sin^{-1}(2x - 1)]$ (2)
- ii) Hence, deduce that $\int_{\frac{3}{4}}^1 \frac{dx}{\sqrt{x-x^2}} = \frac{\pi}{3}$ (2)

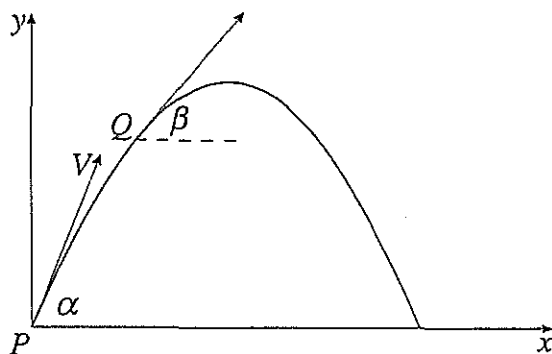
Question 7 (Start a new page).**(12 marks).**

- a) The rate at which a body cools is assumed to be proportional to the difference between its temperature T and the constant temperature P of the surrounding medium.

This can be expressed by the differential equation:

$$\frac{dT}{dt} = k(T - P) \quad \text{where } t \text{ is the time in hours and } k \text{ is a constant.}$$

- i) Show that $T = P + Ae^{kt}$, where A is a constant, is a solution of the differential equation. (1)
- ii) A heated body cools from 90°C to 50°C in 3 hours. The temperature of the surrounding medium is 20°C . Find the temperature of the body after one further hour has elapsed. Give your answer correct to the nearest degree. (3)
- b) A particle is projected from a point P on horizontal ground, with speed $V\text{ms}^{-1}$ at an angle of elevation to the horizontal of α .



Its equations of motion are $\ddot{x} = 0$, $\ddot{y} = -g$.

- i) Write down expressions for its horizontal (x) and vertical (y) displacements from P after t seconds. (2)
- ii) Determine the time of flight of the particle. (1)
- iii) The particle reaches a point Q , as shown, where the direction of flight makes an angle β with the horizontal. Show that the time taken to travel from P to Q is: (2)
- $$\frac{V \sin(\alpha - \beta)}{g \cos \beta} \text{ seconds.}$$
- iv) Consider the case when $\beta = \frac{\alpha}{2}$. If the time taken to travel from P to Q is then one-third of the total time of flight, find the value of α . (3)

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a \neq 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE: $\ln x = \log_e x, x > 0$

Q1 a) $(-3, 5) (8, 4)$ $l:3$
 $m:n$

$$x = \frac{3 \times -3 + 1 \times 8}{4} = -\frac{1}{4}$$

$$y = \frac{3 \times 5 + 1 \times 4}{4} = \frac{19}{4}$$

$$\left(-\frac{1}{4}, \frac{19}{4}\right)$$

b) $y = 8 - 3x$, $m_1 = -3$

$$\tan 45^\circ = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$1 = \left| \frac{-3 - m_2}{1 - 3m_2} \right|$$

$$\frac{-3 - m_2}{1 - 3m_2} = 1$$

or

$$\frac{-3 - m_2}{1 - 3m_2} = -1$$

$$-3 - m_2 = 1 - 3m_2$$

$$2m_2 = 4$$

$$m_2 = 2$$

or

$$-3 - m_2 = 3m_2 - 1$$

$$-2 = 4m_2$$

$$m_2 = -\frac{1}{2}$$

At $(4, 0)$ $y - 0 = 2(x - 4)$
 $y = 2x - 8$

$y = -\frac{1}{2}(x - 4)$
 $y = -\frac{1}{2}x + 2$

c) $\int x \sqrt{3-x} dx$ $u = 3-x$ so $x = 3-u$

$$\frac{du}{dx} = -1 \text{ so } dx = -du$$

$$= \int u^{\frac{1}{2}} (3-u) (-du)$$

$$= \int -3u^{\frac{1}{2}} + u^{\frac{3}{2}} du = \frac{2}{5} u^{\frac{5}{2}} - 2u^{\frac{3}{2}} + C$$

$$= \frac{2}{5} \sqrt{(3-x)^5} - 2\sqrt{(3-x)^3} + C$$

d) $\frac{2x+5}{x} \geq 1$

Case 1 $x > 0$

$$2x+5 \geq x$$

$$x \geq -5$$

$$\therefore x > 0 \rightarrow \text{or}$$

Case 2 $x < 0$

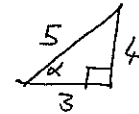
$$x \leq -5$$

$$\therefore x \leq -5$$

1e) $\sin \left[\cos^{-1} \left(\frac{3}{5} \right) + \tan^{-1} \left(-\frac{3}{4} \right) \right] = \frac{1}{25}$ (1)

Let $\alpha = \cos^{-1} \left(\frac{3}{5} \right)$

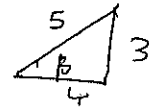
$$\sin \alpha = \frac{4}{5}$$



Let $\beta = \tan^{-1} \left(-\frac{3}{4} \right)$ $-\frac{\pi}{2} < \beta < \frac{\pi}{2}$ Range of $\tan^{-1}$
 $\therefore$ 4th quad

$$\cos \beta = \frac{4}{5}$$

$$\sin \beta = -\frac{3}{5}$$



LHS of (1) $= \sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$
 $= \frac{4}{5} \times \frac{4}{5} + \frac{3}{5} \times -\frac{3}{5} = \frac{7}{25}$

Q5) a) $\int_0^{\frac{\pi}{2}} \cos^2 3x \, dx$

$$= \int_0^{\frac{\pi}{2}} \frac{1}{2} (1 + \cos 6x) \, dx$$

$$= \frac{1}{2} \left[x + \frac{\sin 6x}{6} \right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{2} \left[\left(\frac{\pi}{2} + \frac{\sin 3\pi}{6} \right) - 0 \right]$$

$$= \frac{1}{2} \left[\frac{\pi}{2} + 0 \right]$$

$$= \frac{\pi}{4}$$

b) $-\sin x - \sqrt{3} \cos x$

$$R \cos(x - \alpha) = R(\cos x \cos \alpha + \sin x \sin \alpha)$$

$$R = \sqrt{1^2 + \sqrt{3}^2} = \sqrt{4} = 2$$

$$\therefore \frac{-\sqrt{3} \cos x - \sin x}{2} = \cos x \cos \alpha + \sin x \sin \alpha$$

$$\text{Where } \cos \alpha = \frac{-\sqrt{3}}{2} \quad \sin \alpha = -\frac{1}{2}$$

$$\tan \alpha = \frac{-\frac{1}{2}}{-\frac{\sqrt{3}}{2}} = \frac{1}{\sqrt{3}} \quad \text{3rd quad.}$$

$$\alpha = \frac{7\pi}{6}$$

$$\therefore -\sin x - \sqrt{3} \cos x = 2 \cos \left(x - \frac{7\pi}{6} \right)$$

(OR) $= -2 \cos \left(x - \frac{\pi}{6} \right)$

ii) $2 \cos \left(x - \frac{7\pi}{6} \right) = 2$

$$\cos \left(x - \frac{7\pi}{6} \right) = 1$$

$$x - \frac{7\pi}{6} = 0, 2\pi$$

$$x = 0 + \frac{7\pi}{6}, 2\pi + \frac{7\pi}{6}$$

$$\therefore x = \frac{7\pi}{6}, 0 \leq x \leq 2\pi$$

$$= 3.665$$

c) $\frac{1 - \tan \theta \tan 2\theta}{1 + \tan \theta \tan 2\theta}$

$$= \frac{1 - \tan \theta \left(\frac{2 \tan \theta}{1 - \tan^2 \theta} \right)}{1 + \tan \theta \left(\frac{2 \tan \theta}{1 - \tan^2 \theta} \right)}$$

$$= \frac{1 - \tan^2 \theta - 2 \tan^2 \theta}{1 - \tan^2 \theta + 2 \tan^2 \theta}$$

$$= \frac{1 - 3 \tan^2 \theta}{1 + \tan^2 \theta}$$

$$= \frac{1 - \frac{3 \sin^2 \theta}{\cos^2 \theta}}{1 + \frac{\sin^2 \theta}{\cos^2 \theta}}$$

$$= \frac{\cos^2 \theta - 3 \sin^2 \theta}{\cos^2 \theta + \sin^2 \theta}$$

$$= \cos^2 \theta - 3(1 - \cos^2 \theta)$$

$$= 4 \cos^2 \theta - 3$$

Q6

a) i) $f(x) = x - \frac{1}{x} \quad x > 0$

$$f(x) = x - x^{-1}$$

$$f'(x) = 1 + x^{-2} = 1 + \frac{1}{x^2}$$

$$f'(x) = 0 \quad \leftarrow \text{stationary pts.}$$

$$1 + \frac{1}{x^2} = 0$$

$$\frac{1}{x^2} = -1$$

$$1 = -x^2$$

$$-1 = x^2$$

as $x^2 \neq -1$

$\therefore$ no solution

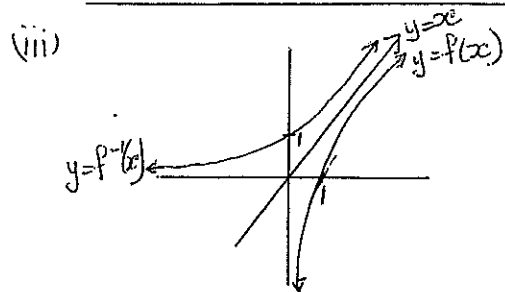
$\therefore$ no stationary points

(ii) As $x \rightarrow \infty \quad \frac{1}{x} \rightarrow 0$

$$\therefore x - \frac{1}{x} \rightarrow x \quad (\text{from below})$$

As $x \rightarrow 0 \quad \frac{1}{x} \rightarrow \infty$

$$\therefore x - \frac{1}{x} \rightarrow -\infty$$



(iv) $x = y - \frac{1}{y}$

$$x + \sqrt{x^2 + 4} = y - \frac{1}{y} + \sqrt{\left(y - \frac{1}{y}\right)^2 + 4}$$

$$= y - \frac{1}{y} + \sqrt{y^2 - 2 + \frac{1}{y^2} + 4}$$

$$= y - \frac{1}{y} + \sqrt{y^2 + 2 + \frac{1}{y^2}}$$

(iv)

$$= y - \frac{1}{y} + \sqrt{\left(y - \frac{1}{y}\right)^2}$$

$$= y - \frac{1}{y} + y + \frac{1}{y}$$

$$= 2y$$

v) $f(x): y = x - \frac{1}{x}$

$$f^{-1}(x): x = y - \frac{1}{y}$$

$$\therefore 2y = x + \sqrt{x^2 + 4}$$

$$y = \frac{x + \sqrt{x^2 + 4}}{2}$$

b) i) $\frac{d}{dx} [\sin^{-1}(2x-1)]$

let $u = 2x-1$

$$= \frac{d}{dx} [\sin^{-1} u] \text{ where } u = 2x-1$$

$$= \frac{1}{\sqrt{1-u^2}} \times \frac{d}{dx} (2x-1)$$

$$= \frac{1}{\sqrt{1-(2x-1)^2}} \times 2$$

$$= \frac{2}{\sqrt{1-(4x^2-4x+1)}}$$

$$= \frac{2}{\sqrt{4x^2-4x}}$$

$$= \frac{2}{\sqrt{4x(x-1)}} = \frac{2}{\sqrt{4(x-x^2)}}$$

$$= \frac{1}{\sqrt{x-x^2}}$$

(ii) $\int_{\frac{3}{2}}^1 \frac{dx}{\sqrt{x-x^2}} = \left[\sin^{-1}(2x-1) \right]_{\frac{3}{2}}^1$

$$= \sin^{-1}(1) - \sin^{-1}\left(\frac{3}{2} - 1\right)$$

$$= \frac{\pi}{2} - \frac{\pi}{6}$$

$$= \frac{\pi}{3}$$

$\therefore \text{RHS.}$

7. a) $T = P + Ae^{kt}$ so $Ae^{kt} = T - P$ ①

$$\frac{dT}{dt} = Ae^{kt} \times k$$

$$= k(T - P) \text{ from ①}$$

$\therefore T = P + Ae^{kt}$ a solution of $\frac{dT}{dt} = k(T - P)$

(ii) $t=0, T=20$ $t=3, T=50$ $P=20$
Find T when $t=4$.

$$T = P + Ae^{kt}$$

$t=0, T=20, 20 = 20 + Ae^0$
 $\therefore A=0$

$\therefore T = 20 + 0e^{kt}$

$t=3, T=50 \therefore 50 = 20 + 70e^{3k}$
 $30 = 70e^{3k}$
 $\frac{3}{7} = e^{3k}$

$\ln(\frac{3}{7}) = 3k$
 $\therefore k = \frac{1}{3} \ln(\frac{3}{7})$ $\therefore T = 20 + 70e^{\frac{1}{3} \ln(\frac{3}{7}) \times 4}$

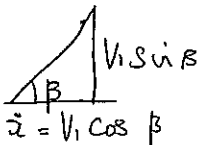
$t=4$ $T = 20 + 70e^{\frac{4}{3} \ln(\frac{3}{7})}$
 $T = 43^\circ$ (to nearest degree)

b) (i) $\ddot{x} = 0$
 $\dot{x} = C_1$
 $t=0, \dot{x} = V \cos \alpha \therefore C_1 = V \cos \alpha$
 $\therefore \dot{x} = V \cos \alpha$
 $x = Vt \cos \alpha + C_2$
 $t=0, x=0 \therefore C_2 = 0$
 $\therefore x = Vt \cos \alpha$

$\ddot{y} = -g$
 $\dot{y} = -gt + C_3$
 $t=0, \dot{y} = V \sin \alpha$
 $\therefore C_3 = V \sin \alpha$
 $\therefore \dot{y} = -gt + V \sin \alpha$
 $y = -\frac{gt^2}{2} + Vt \sin \alpha + C_4$
 $t=0, y=0, \therefore C_4 = 0$
 $\therefore y = -\frac{gt^2}{2} + Vt \sin \alpha$

(ii) Time of flight $y=0$
 $\therefore t(V \sin \alpha - \frac{gt}{2}) = 0$ $t=0$ (start)
 $\therefore V \sin \alpha = \frac{gt}{2} \therefore t = \frac{2V \sin \alpha}{g}$

(iii)



$\tan \beta = \frac{V \sin \alpha}{V \cos \alpha} = \frac{\dot{y}}{\dot{x}}$

Also $\frac{\dot{y}}{\dot{x}} = \frac{V \sin \alpha - gt}{V \cos \alpha}$ where t is time from P to Q .

$\therefore \frac{\sin \beta}{\cos \beta} = \frac{V \sin \alpha - gt}{V \cos \alpha}$

$V \sin \beta \cos \alpha = \cos \beta (V \sin \alpha - gt)$

$V \sin \beta \cos \alpha = V \sin \alpha \cos \beta - gt \cos \beta$

$\therefore gt \cos \beta = V \sin \alpha \cos \beta - V \cos \alpha \sin \beta$
 $= V \sin(\alpha - \beta)$

$\therefore t = \frac{V \sin(\alpha - \beta)}{g \cos \beta}$

(iv) $\frac{V \sin(\alpha - \beta)}{g \cos \beta} = \frac{1}{3} \left(\frac{2V \sin \alpha}{g} \right) \rightarrow \text{time of flight}$

$\beta = \frac{\alpha}{2} \therefore \frac{V \sin(\alpha - \frac{\alpha}{2})}{g \cos \frac{\alpha}{2}} = \frac{2}{3} \frac{V \sin \alpha}{g}$

$\frac{\sin \frac{\alpha}{2}}{\cos \frac{\alpha}{2}} = \frac{2}{3} \sin \alpha$

$\frac{\sin \frac{\alpha}{2}}{\cos \frac{\alpha}{2}} = \frac{2}{3} (2 \sin \frac{\alpha}{2} \cos \frac{\alpha}{2})$ double angle

$3 \sin \frac{\alpha}{2} = 4 \sin \frac{\alpha}{2} \cos \frac{\alpha}{2} \cos \frac{\alpha}{2}$
 $4 \sin \frac{\alpha}{2} \cos^2 \frac{\alpha}{2} - 3 \sin \frac{\alpha}{2} = 0$

$\sin \frac{\alpha}{2} (4 \cos^2 \frac{\alpha}{2} - 3) = 0$

$\sin \frac{\alpha}{2} = 0$ $\frac{\alpha}{2} = 0 \times$

$\therefore \cos^2 \frac{\alpha}{2} = \frac{3}{4}$ $\cos \frac{\alpha}{2} = \frac{\sqrt{3}}{2}$ $\frac{\alpha}{2} = \frac{\pi}{6}$
 $\alpha = \frac{\pi}{3}$

